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The Transmission of Home Garden Knowledge: Safeguarding Biocultural Diversity and Enhancing Social–Ecological Resilience

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ABSTRACT
The last decades have witnessed a growing research interest of local ecological knowledge (LEK), with some research focusing on its effective transmission for natural resource management. Here we contribute to this body of research by focusing on an understudied agroecosystem: home gardens in rural areas of developed countries. We characterize home garden knowledge in Vall de Gósol (Catalan Pyrenees) and analyze the modes of transmission of such knowledge to discuss how such mechanisms might affect home garden resilience. We identify a diverse local home garden knowledge, which is mainly transmitted from parents to child. Members of the parental generation other than the parents and individuals of the same generation were only important for the transmission of some specific knowledge. We conclude that home gardens are biocultural refugia in a world of decreasing complex local knowledge systems and that different cultural transmission modes confer diversity and enhance social–ecological resilience in those agroecosystems.

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During the 1992 Convention on Biological Diversity, the international scientific community acknowledged the indivisible link between cultural and biological diversity and the relevance of local ecological knowledge (*sensu* Berkes[1]) for their conservation (Maxted et al. 2002). Since then, research interest has grown to deepen our understanding of the role of local ecological knowledge (hereafter LEK) in the effective management of natural resources and in the protection of biological and cultural diversity (hereafter biocultural diversity), (e.g., Berkes, Colding, and Folke 2000; Calvet-Mir 2011; Barthel, Crumley, and Svedin 2013).

Some authors have highlighted the applicability of LEK, arguing that such knowledge systems can guide decision making for sustainable management of natural resources (Berkes and Folke 2002). Others have shown that local experts are able to understand and deal with multiscalar and complex social–ecological processes by intuitively using

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Color versions of one or more of the figures in the article can be found online at www.tandfonline.com/usnr.

[1]Local ecological knowledge refers to the cumulative body of knowledge, practices, and beliefs about local ecosystems and their management evolving by adaptive processes and shared among generations by cultural transmission (adapted from Berkes, Colding, and Folke 2000, 1252).

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insights from the historical perspective encoded in LEK systems (Chalmers and Fabricius 2007). Particularly, in situations of change or when uncertainty is high, LEK can provide alternative sources of coping and adaptive strategies (Berkes, Colding, and Folke 2000; Reyes-García, Salpeteur, and Calvet-Mir et al. 2014).

Due to the potential of LEK for biodiversity conservation, management of natural resources, and adaptation to global change, researchers have shown interest in understanding how the information encapsulated in local knowledge is generated, transmitted, adapted, and lost (Atran, Medin, and Ross 2004; Lozada, Ladio, and Weigandt 2006; Reyes-García et al. 2013). While research on the dynamics of LEK has been dominated by case studies in indigenous communities (Ohmagari and Berkes 1997; Demps et al. 2012), it is still scarce in nonindigenous rural areas where such bodies of knowledge are also present (for a state of the art in Europe see Hernández-Morcillo et al. 2014). Such research is urgent in rural settings in developed countries, where recent demographic, economic, and cultural changes (i.e., reduction in the number of farms, migration to urban areas, agricultural incentives, globalization, and simplification of diets) threaten the maintenance of diversified agroecosystems (Naredo 2004) and reduce complex local knowledge systems into "pockets of LEK" (Gómez-Baggethun et al. 2010).

Aiming to contribute to such research, here we analyze LEK associated with home gardens in Vall de Gósol, Catalan Pyrenees. We focus on home gardens because previous research has highlighted their importance in the maintenance of LEK in developed countries (Vogl, Axmann, and Vogl-Lukasser 2004; Galluzzi, Eyzaguirre, and Negri 2010; Calvet-Mir, Gómez-Bagetthun, and Reyes-García 2012). We define "home garden" as a small, fenced plot close to a gardener's homestead, where annual, biennial, and/or perennial cultivated species are grown in beds (Vogl and Vogl-Lukasser 2003). Home gardens can be seen as biocultural conservation agents (Calvet-Mir 2011) or biocultural refugia (Barthel, Crumley, and Svedin 2013) contributing to the conservation of both biological and cultural diversity.

In what follows, we present existing evidence on LEK transmission. We then characterize home garden knowledge in the study area and assess how home garden knowledge is transmitted among Vall de Gósol gardeners. Other than exemplifying the persistence of pockets of LEK in rural areas of the developed world, this work helps stress the role of site-specific experiences in storing and transmitting knowledge for ecosystems management (e.g., Berkes, Colding, and Folke 2003; Barthel, Folke, and Colding 2010). Understanding the modes of transmission of knowledge that guides the management of species, habitats and other agroecosystems features, and particularly home gardens, can help us appreciate the central role local knowledge systems play in social–ecological resilience (Calvet-Mir 2011; Barthel, Crumley, and Svedin 2013).

The Transmission of Local Ecological Knowledge

Researchers differentiate between processes of social learning and modes of cultural transmission (Hewlett et al. 2011). Processes of social learning include the mechanisms through which knowledge is acquired (i.e., own experimentation, teaching, emulation, imitation, and collaborative learning; Hewlett et al. 2011). Here we focus on teaching as a process of information transmission on home garden management and specific crop uses. Home garden informants might be more aware of the importance of particular "teachers" than

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of the complexity entailed in other social learning processes (Reyes-García, Gallois, and Demps 2015).

Modes of transmission refer to individuals from whom to learn (i.e., parents, elders, peers, the prestigious, the majority). Based on the work of Cavalli-Sforza and Feldman (1981), social anthropologists conducting empirical research on the transmission of cultural traits have studied three nonexclusive modes in which knowledge is transmitted (i.e., from whom people learn): (1) vertical transmission, that is, from the parental generation to the offspring generations; (2) horizontal transmission, that is, between individuals of the same generation; and (3) oblique transmission, that is, from members of the parental generation other than the parents. Sociologists (Bengtson and Troll 1978) and psychologists (Schönpflug 2008) provided arguments demonstrating that transmission is not a one-way street. For example, some authors characterize the transmission of knowledge as a transactional process in which the parent impacts the child, but in which the child might also influence the parent, a phenomenon known as retroactive transmission (Pinquart and Silbereisen 2004). Boyd and Richerson (1985) and Henrich and McElreath (2003) expanded the field beyond the direction of knowledge transmission by modeling other nonvertical transmission mechanisms such as conformist transmission, that is, copying the beliefs or practices of the majority, or prestige bias, that is, copying the successful.

Although several anthropologists have argued that LEK is mainly transmitted vertically (Lancy 1999; Hewlett, De Silvestri, and Guglielmino 2002), other works showed that transmission modes are context dependent (Hewlett et al. 2011; Demps et al. 2012). That is, depending on the type of knowledge or skills to be transmitted, or on the age in which knowledge is acquired, other modes of transmission might be dominant. The importance of vertical transmission has thus been overrated (Aunger 2000), although it is expected that neither oblique nor horizontal transmission dominates across all cultural domains (McElreath and Strimling 2008).

Despite the proliferation of literature in the fields of LEK and its transmission among rural and indigenous societies, there are important gaps in the research in the context of developed countries. As LEK is mainly built, assimilated, and spread in the natural environment (Atran, Medin, and Ross 2004), the study of LEK transmission in remaining home gardens in developed settings, where severe deruralization processes have occurred in the last decades, is relevant to avoid the loss of this knowledge.

Methods

Study Site

Vall de Gósol is the name of a valley situated at the foot of Pedraforca's massif, in the central Catalan Pyrenees, among Cadí, Verd, and Ensija mountains. The valley is approximately 20 km^2 and includes 4 villages, taking the name of the largest of them, Gósol (Figure 1). Around 220 inhabitants live in these four villages, although most of them are concentrated in Gósol (198 inhabitants).

Customarily, men in the area worked as sheep or cattle dealers, often spending parts of the year out of the valley working as agricultural wage laborers. Women were in charge of household chores including home garden tasks, although some tasks were mainly performed by men (e.g., fertilization and disease management). Traditional division of work

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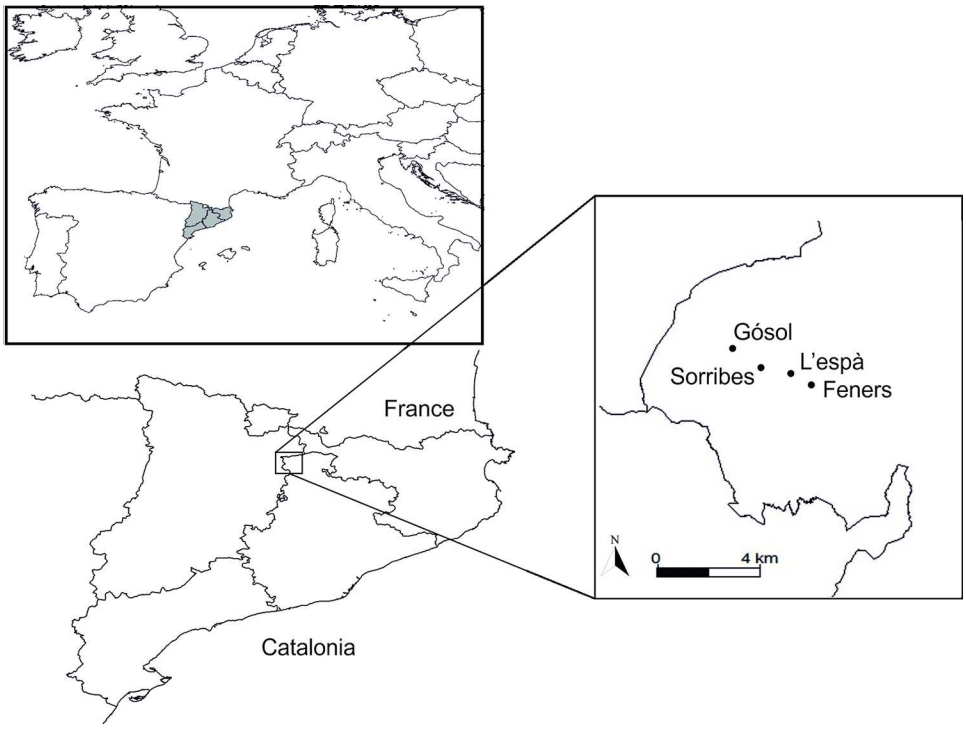

Figure 1. Map of the study area.

has been recently reversed, since men manage 71% of the home gardens studied (Riu-Bosoms, Calvet-Mir, and Reyes-García 2014). Such a trend is explained by overall changes in the area that parallel demographic, economic, and cultural changes found in other rural areas of the Iberian Peninsula (Naredo 2004). Since the 1960s, subsistence agriculture has been progressively abandoned due to the introduction of the commercial dairy sector in the area. Soon after, at the end of the 1970s, the opening of a road that connected previously isolated villages with the local market town (Berga) resulted in a complete integration of the local economic system into the market economy. The majority of farms were no longer multifunctional and they specialized their production, increasing their market dependence. The valley's population experienced a steep decline, as many families decided to leave the valley in search of employment opportunities in the Barcelona metropolitan area. Thus, in the last 60 years, the number of people living in the area has decreased by about 60%, basically due to urban migration (INE 2015), resulting in the decline of the number of farms. Currently, only 10% of the population is devoted to agrarian activities (IDESCAT 2015), whereas there is a clear predominance of the service sector, with an occupancy rate of 60% because people combine agricultural activities with tourist services. The industry sector employs 18% of the population, while 12% work in the construction sector.

As a consequence of the geographic isolation of the area, home gardens played a key role in the subsistence of local population for many years. Currently, they are still deeply integrated in the valley's day-to-day life, although nowadays gardeners do not depend on home gardens for subsistence. Home gardens have an average size of 202 m^2, ranging from 25 to 711 m^2. In a previous work, we found 10 crops that conformed to the definition of landrace,

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a high number for the relatively small geographical area studied, which highlights the importance of home gardens to preserving the agricultural genetic pool in Vall de Gósol (Riu-Bosoms, Calvet-Mir, and Reyes-García 2014).

Sampling

We conducted research in the four villages that are geographically within Vall de Gósol and that share ecological and cultural characteristics. We did not extend the sample outside the valley because there are substantial differences in the environmental conditions and dialectal variations of the local language, Catalan. We collected data with two different samples. First, to describe and characterize home garden knowledge, we followed Calvet-Mir et al. (2011) and chose experienced gardeners (or people who have managed a home garden as primary gardener for more than 25 years). We found seven people (four women and three men) with such characteristics. The selected gardeners were aged between 71 and 88 years. Second, to assess the prevalence of modes of transmission of home garden knowledge, we interviewed all the people in the valley currently managing a home garden. In total, we interviewed 24 gardeners, 7 women and 17 men, aged between 21 and 85 years.

Data Collection

We collected data between March and July 2011 and between January and June 2012 through mixed methods.

We used participant and nonparticipant observation throughout all the research period to achieve a better understanding of knowledge, practices, and beliefs associated with home gardens management. Data were collected through annotations. The second author of this article, a resident of the valley, managed a home garden under the instruction of an elder gardener. In addition, during fieldwork we occasionally helped gardeners in daily tasks such as planting and harvesting. Interactions in these informal occasions helped us to gain a deeper understanding of gardeners' knowledge and skills to manage a home garden.

In 2011 we conducted semistructured interviews with the experienced gardeners, asking them about the origin of crop seeds, crop distribution in the home garden, crop uses, and specific characteristics of each crop (i.e., soil preference, seed selection criteria, seasonality). We also asked about sayings and proverbs associated with crops and home gardens management, weather forecasts, and home garden beliefs. Information from semistructured interviews was mostly noted down during interviews.

During January and June 2012 we conducted a survey to assess (1) the individual knowledge related to home garden management and (2) the gardener's main teacher, according to the gardener's perception. The survey was constructed using information gathered in semistructured interviews (see also Riu-Bosoms, Calvet-Mir, and Reyes-García 2014). We first presented respondents with seven hypothetical situations covering different fields of home garden knowledge (e.g., sowing and harvesting a specific crop). We then asked the informant how she or he would deal with the situation. The question mobilized information on skills, practices, knowledge, and beliefs that the informant would use to manage this situation. After that, we asked about the main source of information for the answers provided. For example, after the informant had presented the most appropriate fertilization technique for black pea (*Pisum sativum* L.) landrace, we asked the person to name the person who

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taught him or her about this technique. Finally, we collected sociodemographic information on the gardeners (i.e., age, years living in the valley, years tending a garden).

In June 2012 we organized a participatory workshop with eight gardeners. The second author as facilitator of the workshop engaged participants in a dialogue and discussion to reflect about the topics of the semistructured interviews (Steiner 1999). We recorded the workshop and transcribed participants' contributions to validate the information obtained in semistructured interviews and to avoid recall bias (i.e., the differences in the accuracy or completeness of the recollections retrieved by study participants regarding events or experiences from the past).

Data Analysis

We coded data gathered through participant and nonparticipant observation, semistructured interviews, and the participatory workshop by using six categories, not predefined, that emerged across an inductive process (Newing 2011), which related to their knowledge on home garden management: (1) crop sowing and harvesting; (2) water management; (3) fertilization; (4) pest and disease management; (5) seed selection and storage; and (6) crop uses. For each management category we selected an example of the most common cultural display (i.e., sayings, predictions, specific management techniques, recipes, and festivities).

We then drew on the literature on modes of cultural transmission to classify sources of information from each particular situation presented in the survey (Cavalli-Sforza and Feldman 1981; Hewlett et al. 2011). Verbatim answers on sources of information for each management practice were classified as (1) parents; (2) older kin and non-kin (excluding parents); (3) own cohort; and (4) younger cohort. We used descriptive statistics to explore the sources of home garden management knowledge. We first used background data to build a general model that we graphically represented with a network diagram. We then ran descriptive statistics on the source of information for the seven hypothetical situations in our survey. These situations match with the six categories created for classifying home garden knowledge, but we include another situation to cover a new form of knowledge present in home gardens that comes from new technologies.

Results

Local Ecological Knowledge in Vall De Gósol Home Gardens

Our results show that there is a rich and diverse knowledge related to both the general management of home gardens and the management and uses of specific crops grown in home gardens (Figure 2). We found a total of 55 sayings, predictions, practices, recipes, and others cultural manifestations related to home garden knowledge. In this section, we present one example of a common cultural display for each management category so to exemplify the diversity of home garden knowledge.

Crop Sowing and Harvesting

The sowing and harvesting calendar is usually expressed through oral sayings relying upon the Catholic and the lunar calendar. For example, the sowing of garlic (*all, Allium*

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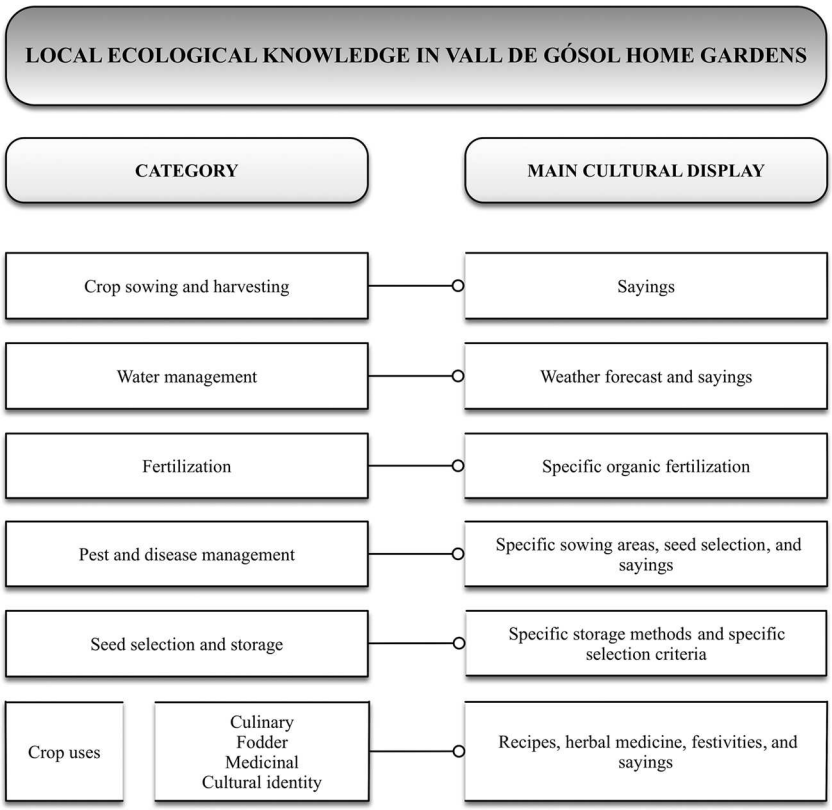

Figure 2. Management categories and most common cultural display of local ecological knowledge in Vall de Gósol home gardens.

sativum L.) is marked by both Catholic festivities and individual organoleptic preferences, as gardeners argue that garlic taste changes according to the time of sowing. According to a local saying: "If you want a good garlic harvest, sow it during the old moon of January [the period that goes from full to new moon], if you want a good and mellow garlic, sow it during Saint Martin [November eleventh], and if you want a pungent garlic, sow it during advent time [December]."

Water Management

Local knowledge about water management includes weather forecast based on two types of indicators: (1) elements of the local geography and (2) animal behavior. For example, according to local tradition, the presence of clouds on "*Els Tres Collets*" (local toponym of a mountain) at mid-morning time indicates scattered rainfall in the afternoon of the same day, whereas a cloud accumulation looking like a castle on top of "*El Verd*" (local toponym of another mountain) indicates a generalized rain over the valley. Animal behavior has also been used as indicator to predict rainfall, as it is reflected in a saying according to which a specific cuckoo singing predicts the arrival of rains. Such observations guide local decisions on the irrigation of home gardens and fields.

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Fertilization

Organic fertilization is a traditional and generalized practice in the valley. Crops are fertilized using livestock and hens' dung, ashes, or other natural fertilizers. According to local knowledge, each type of crop requires a different fertilizer, in specific doses, that should be applied using specific techniques. Furthermore, to ensure proper supply of nutrients, gardeners must also make crop rotations. For instance, a landrace of black pea (*pèsol negre, Pisum sativum* L.) is not fertilized and is sown in sandy soil, rather poor in nutrients, to avoid the plant growing very tall and lying on the ground. According to gardeners, this avoids the preparation of trellises, without affecting pea production. On the contrary, cabbage (*col d'hivern, Brassica oleracea* L. var. *oleracea*) needs to be fertilized often for adequate yields, preferably with hens' manure, which is a very concentrated manure that is not considered suitable for all crops.

Pest and Disease Management

Gardeners also manage their crops so to reduce the chances of crop failure. For example, gardeners plant potatoes in fields located in areas of high elevation. Subsequently, they select seed potatoes from those fields, which they plant in lower elevation fields where the so-selected seeds would have a higher yield and would also be more convenient to harvest. According to local gardeners, this technique prevents diseases, as at high altitudes potatoes are less likely to suffer diseases. Differently, and still according to gardeners, cereals' yields increase if the seeds are sowed at altitudes higher than the altitude of the place where they were produced. Such practices are encoded in a local saying: "Barley wants to go up and potato wants to go down."

Seed Selection and Storage

Gardeners use specific methods and criteria to select and store seeds for different crops. In the case of garlic, the common practice is to make a garlic braid of 100 bulbs (locally known as "*forc*"). Gardeners hang garlic braids in a cool and dry place to preserve them for later consumption or sowing. Garlic seed selection depends on size, color, and situation into the bulb. For example, to conserve the pungent qualities of garlic, the selected seed must be big, reddish, and placed in the external part of the bulb.

Crop Uses

While some crops are mainly used for culinary purposes, others are multipurpose. For example, a local landrace of cabbage is harvested for human consumption only after the first frost, although its leaves are harvested before, as fodder for hens and rabbits. The same crop can also be used as a medicinal, to diminish stomachache. Nowadays some species that used to play an important role in daily diet in the past are only grown for cultural reasons. For example, this is the case of a wheat landrace (*forment, Triticum aestivum* L.) that is sowed to be harvested and threshed using traditional tools and methods in the context of the festivity of "Harvest and Threshing" that takes place on the first Sunday of August.

Modes of Transmission of Home Garden Knowledge

Survey data suggest that gardeners acquire home garden knowledge mainly from parents (i.e., vertical transmission), first father and then mother, together representing about 50% of the responses (Figure 3). They also learn from members of the parental generation other than the parents (i.e., oblique transmission), representing about 30% of the responses, and from individuals of the same generation (i.e., horizontal transmission). Results also show another mode of transmission: from members of younger to members of older generations, which we call here "retroactive transmission," following the work of Bengtson and Troll (1978), who used the term to refer to child-to-parent transmission. Horizontal and retroactive modes of transmission are minor compared with the previous ones, but very important for some specific bodies of knowledge, as described in the following.

Vertical, horizontal, and oblique transmissions are represented in five of the six categories analyzed in Figure 2—that is, in all except "pest and disease management," for which horizontal transmission was not reported. Vertical and oblique transmissions are the main modes of cultural transmission in all categories (Table 1). For "crop sowing and harvesting," vertical transmission represents 72.72% of the cases. "Water management" is mainly obliquely transmitted (53.85%). The transmission of fertilization knowledge is principally performed by vertical transmission (73.33%). In this category, horizontal transmission also takes a relevant role (20%). "Pest and disease management" is acquired via both vertical (52.94%) and oblique (47.06%) transmission. Similarly "seed selection and storage" is acquired through vertical (36.84%) and oblique transmission (52.63%). Lastly, the knowledge embedded in the "crop uses" category mainly comes from vertical (47.82%) and oblique (43.48%) transmission.

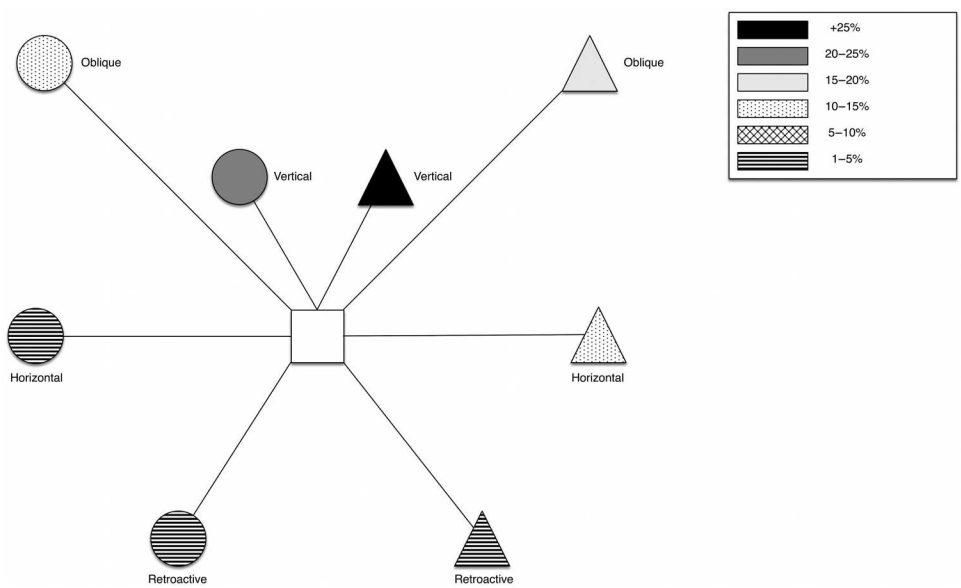

Figure 3. Modes of transmission of home garden knowledge in Vall de Gósol. *Note*: Nodes (gardener and the different sources of knowledge acquisition) are shaped by the sex of the node (triangle for men, circle for women, and a square for the gardener, since gardener includes both men and women), and shaded for the percentage of knowledge transmitted by each source. The text next to the node corresponds to the transmission mode.

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Table 1. Percentage of the different modes of knowledge transmission for the different situations presented to gardeners.

Situation	*N* (%)[a]	Vertical transmission (%)	Oblique transmission (%)	Horizontal transmission (%)	Retroactive transmission (%)
Crop sowing and harvesting	22 (91.67)	72.72	9.09	13.63	4.54
Water management	13 (54.17)	38.46	53.85	7.69	0
Fertilization	15 (62.50)	73.33	6.67	20.00	0
Pest and disease management	17 (70.83)	52.94	47.06	0	0
Seed selection and storage	19 (79.17)	36.84	52.63	10.53	0
Crop uses	23 (95.83)	47.82	43.48	8.70	0
New technologies	21 (87.50)	4.76	9.52	57.14	28.57

[a]Gardeners whose response matched information reported in surveys and were asked about the source of transmission.

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Interestingly, new technologies are being integrated into home garden management and seem to be used as coping mechanisms to deal with social–ecological changes. For instance, gardeners reported that black peas (*Pisum sativum* L.) were affected by the pea weevil (*Bruchus pisorum*), a previously unknown pest in the area, which they attribute to the rise of temperatures. To preserve peas from black weevil, once harvested, gardeners stored peas in the freezer for 3 days, avoiding the proliferation of the pest. When we analyzed the modes of transmission of such new knowledge we realized that the patterns were substantially different from the ones previously described, with horizontal transmission having a main role, as about 50% of the knowledge gathered by the gardeners is transmitted by their peers. It should be stressed that about 30% of the people reported as teachers are younger than the informant, highlighting the importance of retroactive transmission. Another distinguishing feature here is the low weight of transmission from older generations, that is, vertical and oblique transmission.

Discussion

We start the discussion by highlighting two limitations of the work presented here, before we discuss our main results. First, the use of fixed categories to classify LEK is artificial, as the successful management of a species depends on information that we have coded under different categories, and partial knowledge (i.e., knowledge regarding a single category) might result in crop failure. Despite acknowledging the holistic nature of LEK, we decided to categorize it to ease our understanding of a complex process. Second, in our interviews, we only recorded the perceived main source of information. It is probable, however, that gardeners combine information from several sources, which likely means that our results underestimate the importance of certain modes of transmission.

Although our case study is limited to a small and rather isolated geographical area, this study is valuable since it provides a first understanding of the relation between cultural transmission and LEK in home gardens in developed countries, a subject not studied before. In that sense, two significant findings of this study stand out. First, results from this work highlight that home gardens in a developed country have associated a rich and diverse pool of LEK; we argue that in such a context LEK is critical in maintaining home gardens as biocultural refugia. Second, we found that home garden knowledge is transmitted through

several sources; we argue that such diversity in sources for the transmission of knowledge might enhance the social–ecological resilience of home gardens.

Maintaining LEK to Preserve Biocultural Diversity in Home Gardens

In accordance with previous research, our results show that in contemporary developed societies, local knowledge related to home gardens (1) helps in home gardens management (Brookfield, Parsons, and Brookfield 2003); (2) provides information on different crop uses (Negri 2003); (3) preserves intra- and interspecific crop genetic diversity (Galluzzi, Eyzaguirre, and Negri 2010); and (4) allows for adaptation to change (Reyes-García, Aceituno-Mata, and Calvet-Mir et al. 2014).

Our results are also consistent with findings from previous research in a geographically and culturally close area (Vall Fosca), where home gardens are a marker of cultural identity (Calvet-Mir et al. 2011; Calvet-Mir, Gómez-Bagetthun, and Reyes-García 2012). Individual sense of place might act as an incentive to conserve landraces in the area (Riu-Bosoms, Calvet-Mir, and Reyes-García 2014), and a motivation for people to preserve the LEK associated to these landraces. According to Clayton (2007), psychological benefits on managing a home garden can be partly attributed to the sense of place. Other authors have also highlighted the importance of LEK to preserve site-specific cultural identity, promoting social cohesion and collective identity (Brookfield, Parsons, and Brookfield 2003). The future trend of such a sense of place among young generations, however, is uncertain. On the one side, the urbanization process, which is pushing young people to migrate to cities in search of employment, might undermine the transmission of home garden knowledge and thus negatively affect the resilience of the system. On the other side, such a sense of place could be strengthened by the growing number of young people who seek to rediscover traditional ways of making a living, which might thus contribute to LEK conservation. In sum, a main finding of our research is that the rich and diverse pool of LEK associated with home gardens remaining in rural areas of developed countries allows such home gardens to become biocultural refugia (Barthel, Crumley, and Svedin 2013). Specific knowledge and experiences about practical management of biodiversity and ecosystem services remain thus embedded in home gardens (Nabhan 2008; Calvet-Mir et al. 2011; Calvet-Mir, Gómez-Bagetthun, and Reyes-García 2012; Barthel, Crumley, and Svedin 2013). Diversity and knowledge encountered and stored in home gardens are part of what has been called social memory (Barthel, Folke, and Colding 2010) or biocultural memory (Toledo and Barrera-Bassols 2008), the present-day expression of a long historical legacy of interrelations between humans and nature, which enhances social–ecological resilience (Nykvist and von Heland 2014).

Cultural Transmission as a Source of Resilience

We have documented modes of transmission of home gardens knowledge in Vall de Gósol. Our results are in agreement with other studies of LEK transmission that suggests that the different features of LEK might explain the existence of different modes of cultural transmission (Aunger 2000). Although vertical transmission appears as dominant (for similar results see Lancy 1999; Hewlett, De Silvestri, and Guglielmino 2002), we also observe that other transmission modes (i.e., oblique and horizontal) are common. Furthermore, our

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data suggest a previously neglected mode of cultural transmission: retroactive transmission, in which information seems to flow from members of younger to members of older generations. This neglected mode of transmission is found in relation to the adoption of new technologies. Technological knowledge is transversal and transmitted through networks of informal contacts between gardeners and seems to promote a type of syncretism with LEK systems, a phenomenon that has also been documented regarding marine (Aswani et al. 2007), forest (Mehring et al. 2011), and agricultural systems (Reyes-García et al. 2013).

We argue that the combination of different modes of transmission of knowledge relevant for the management of home gardens may increase their resilience. Resilience has been defined as the system's ability to recover positively from a perturbation while maintaining social and ecological functions through increasing people's ability to anticipate changes and plan for the future (Adger et al. 2005, 2011; Gómez-Baggethun et al. 2012). In the context of this study, the combination of different transmission modes might confer to home garden managers diversified information, all-important for the adaptive management of social–ecological systems. Such features include social memory, heterogeneity, social learning, and adaptive capacity (Berkes and Folke 1998; Berkes, Colding, and Folke 2000; Bodin, Crona, and Ernstson 2006). Thus, vertical transmission allows for the accumulation of large bodies of LEK, maintaining social memory, or the collective knowledge and experience to be used in times of change and uncertainty. Simultaneous reliance on different modes of transmission (i.e., vertical, oblique, horizontal, and retroactive) ensures flexibility in multiple learning pathways, giving the opportunity to shift learning from one source to another, according to the need. Moreover, simultaneous reliance on different modes of transmission might broaden the collective knowledge base and increase the capacity for innovation and maintenance of different knowledge systems and frameworks for interpretation, thus conferring heterogeneity onto home gardens. Local communities are involved in a process of continuous social learning through the transmission and sharing of knowledge and ideas, and thus resource management can be updated and adapted to changing conditions. Finally, retroactive transmission gives adaptive capacity to home gardens since it is based on the acquisition of new knowledge to cope with changing conditions or innovation to meet new needs.

A recent review on the implications of traditional ecological knowledge for social–ecological resilience in the context of community-based biodiversity conservation initiatives (Ruiz-Mallén and Corbera 2013) highlights the importance of transmission and maintenance of traditional ecological knowledge to cope with disturbances. We hypothesize that the same mechanism might help confer resilience to Vall de Gósol home gardens and it is related to the dynamic nature of LEK. Furthermore, as in community-based conservation contexts, we suggest that, even though current environmental change is faster than the processes of generating and transmitting LEK, gardeners in Vall the Gósol have been able to learn from experience and errors, have engaged young people in learning processes, and have innovated and generated new knowledge to overcome crisis.

Conclusions

Our findings suggest that the maintenance of home gardens and associated LEK is inextricable; moreover, LEK helps transform home gardens from an agricultural system to biocultural refugia. Furthermore, the different pathways through which information on home

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gardens management flows may make up an important factor explaining the rich and diverse body of knowledge present in home gardens, also conferring resilience to those agroecosystems.

As the conservation of LEK has been deemed a priority worldwide, the implications of our findings are not limited to Vall de Gósol home gardens. However, further research is needed to verify how generalizable or widely applicable our findings are. This will require deeply analyzing the dynamics of LEK (creation, acquisition, transformation, transmission, maintenance and loss) and their influence in the social–ecological resilience of home gardens in the current context of urbanization and globalization.

We conclude by raising some environmental policy implications of our work. In developed settings where huge changes in social relations, demography, and culture take place, home gardens might act as important settings to preserve the long historical legacy of interrelations between humans and nature (social memory). Furthermore, home gardens knowledge seems to play a central role in the resilience of the social–ecological systems in which they are embedded. In such situation, it is of paramount importance to support and promote, both informally and officially, the maintenance of home gardens, including rural home gardens and other types of gardens such as urban allotments or community gardens that are nowadays gaining importance in urban planning.

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